

Michael Einhorn

<https://michaeeinhorn.github.io/Portfolio/>

Skills

Python, C, C++, C#, Java, R	Machine Learning, Computer Vision	Data Cleaning and Preprocessing
Julia, SQL, Java Springboot.	Natural Language Processing	Data Visualization in PyPlot and GGPlot
AWS Serverless Development	Finetuning large models with	Parallel Processing with CPU
Linux Shell Scripting	Deepspeed	Multithreading, GPU CUDA, and
Agile and Test Driven Development	PyTorch, Keras, TensorFlow	MPI Distributed Computing

Education

Georgia Institute of Technology, College of Computing, Graduated May 2023

Major: Computer Science, GPA: 3.90

Concentrations: Artificial Intelligence, Modeling and Simulation

Work Experience:

Computer Scientist US Air Force 453rd Electronic Warfare Squadron: Jan 2024 – Present

Modernized legacy Javascript software into a Pub/Sub architecture to automate processes and enable flexibility for future features. Developed modular containerized applications with Python and Docker Swarm. Reduced the time to setup an environment for a new developer from one week to one hour with automated and consistent deployments in Ansible.

Developed software for signal flagging using Java, Springboot and PostgreSQL. Adapted a spellcheck algorithm to detect dropped pulse repetition intervals.

Georgia Tech COAR Lab: Spring 2022 - 2023

Worked with a graduate student to experimentally test their theoretical predictions for the convergence and linear speedup of federated reinforcement learning. Tested Tabular Q-learning, and Convolutional Networks with PPO. Analyzed results in R and wrote an [Undergraduate Thesis](#).

Stanford Existential Risk Initiative Machine Learning Alignment Theory Scholars: Fall 2022

Research Engineer on a multidisciplinary team in Berkeley and London. Finetuned Language Models GPT-Neo 1.3B and GPT-Neox 20B with PPO using Deepspeed with multiple GPUs to play text adventures. Compared different RL algorithms for mode collapse on a toy model playing prisoners' dilemmas and wrote a [post on LessWrong](#) documenting our results.

Orca IoT AWS Developer: Spring and Summer 2021

Developed and code reviewed production software for management of construction sites. Developed Serverless APIs with Python using AWS Lambda, and Cloud Formation. Deployed a Yolo model in AWS Neuron which was both more accurate and less expensive than AWS Rekognition analyzing our in-house dataset. Created a [marketing tech demo](#) for visualizing object detection boxes with real data in the SQL server. Tested models on edge devices such as Coral TPU and Jetson Nano.

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Georgia Tech Research Institute CIPHER: Fall 2021

Developed methods to determine scaling laws for neural networks to predict the amount of data needed and statistically evaluate improvements to scaling laws. Used this procedure to test Resnet with Mixup data augmentation on the CIFAR-10 image dataset. Analyzed data with regression in R using blocking to incorporate results from multiple experiments. Tested for heteroskedasticity and non-normality of residuals and adjusted with Iteratively Reweighted Least Squares. Wrote a draft report on results.

Georgia Tech Research with CHAT (Cetacean Hearing and Telemetry): Spring 2021

Developed an AI application to facilitate real time communication with dolphins. Used Cascade Classifier in OpenCV to detect dolphin whistles in a spectrogram. Tested data preprocessing, and hyperparameters for Precision, Recall, and Inference Time. Debugged multithreaded C code with Valgrind.

Projects:

Blending Music with an Autoencoder

Presented at Georgia Tech Music, Art, and Technology Fair:

Modified an Autoencoder Neural Network that composed music to smoothly blend two songs. Presented at the Georgia Tech Music Art and Technology Fair in March of 2020. Within the latent space, it is possible to take a linear path between any 2 latent vectors that represent songs, and the smooth transition in the latent space translates to a smooth transition in the music.

Computer Systems Mentorship using C:

Participated in a two year mentorship program working on a collaborative case study of Quake 3, an early networked 3D game written in C and capable of running efficiently on even the slowest of computers. Added several Vector and Quaternion functions to handle rotation, interpolation, and projection. The use of Quaternions enabled 3D swinging motion, allowing the player to swing through the level like spiderman. Created a library for each of the vector and quaternion methods with documentation and tutorials so that any team in future years would be able to utilize these tools.

Carnegie Mellon: National High School Game Academy:

Attended a 6-week Computer Science course at Carnegie Mellon for game design. Collaborative teams of 5 people were assigned to remake a classic game with a twist, and then make an original idea in VR. Learned how to maintain focus and motivation within the team as the stress of the deadline approached and how to manage code with multiple people. Learned to use Audacity for sound design, photoshop, and Autodesk Maya, while programming the game with the Unity engine in C#.

ASSIP Aspiring Scientist Summer Internship Program at GMU: Summer 2018

I applied and was accepted to ASSIP and was mentored by Dr. Patrick Gillevet. I worked for 7 weeks in the microbiome analysis center looking for correlations between gut bacteria proteins and bile acids.

Inhibition of the Spread of Antibiotic Resistance by Exposure to Omega 6 Fatty Acid: Spring 2017

Advanced to Virginia State Science and Engineering Fair at Virginia Military Institute.

Presented at an American Society of Microbiology meeting at George Mason University.

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Georgia Institute of Technology CS Coursework:

Natural Language Processing:

Implemented models and algorithms such as LSTMs, Attention, Beam Search, Part of Speech Tagging and Classification. Worked on a team project to use a variation of Tuned Lens to interpret Language Models.

Machine Learning and Intro to AI:

Programmed in Python using NumPy. Implemented models and algorithms such as A* Search, Gaussian Mixture Model, Decision Trees, Reinforcement Learning, Semi-Supervised Learning, and Neural Networks. Worked on a semester team project with Semantic Segmentation using Convolutional Neural Networks. After training on labeled images of pet cats and dogs, the model was able to generalize to segment images of wildlife such as elk, bears, and birds.

Perception and Robotics:

Modeling robot perception and planning using probability theory. Projects include using maximum a posteriori to sort items from sensor readings, and different methods for path planning such as RRT.

Computer Simulation:

Analyzing systems of differential equations, discrete models such as cellular automaton. Semester team project on simulating trading and liquidity providing on the Uniswap decentralized crypto exchange.

High Performance Computing:

Wrote efficient parallel algorithms in Julia. Used SSH and Jupyter Notebooks to run code on the Georgia Tech Pace Computing Cluster. Implemented algorithms with shared memory CPU multithreading, CUDA GPU multithreading, and MPI distributed memory computing. Algorithms include Dense/Sparse Matrix Multiplication, FFT, and sorting.